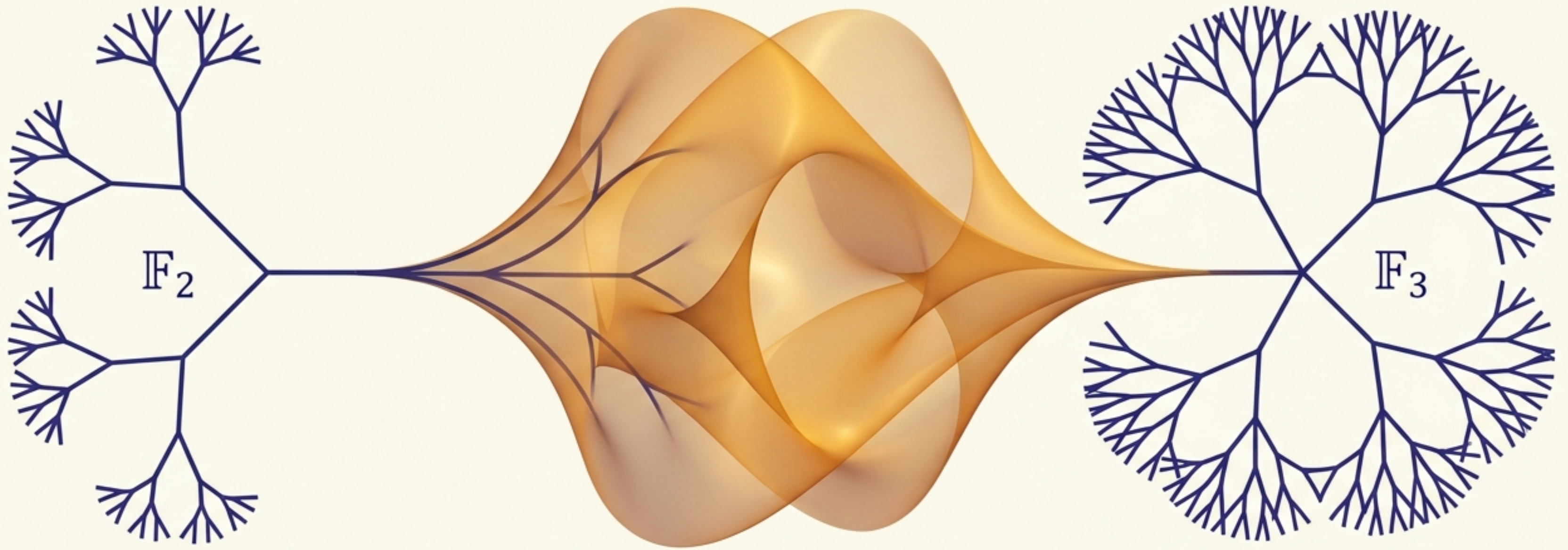
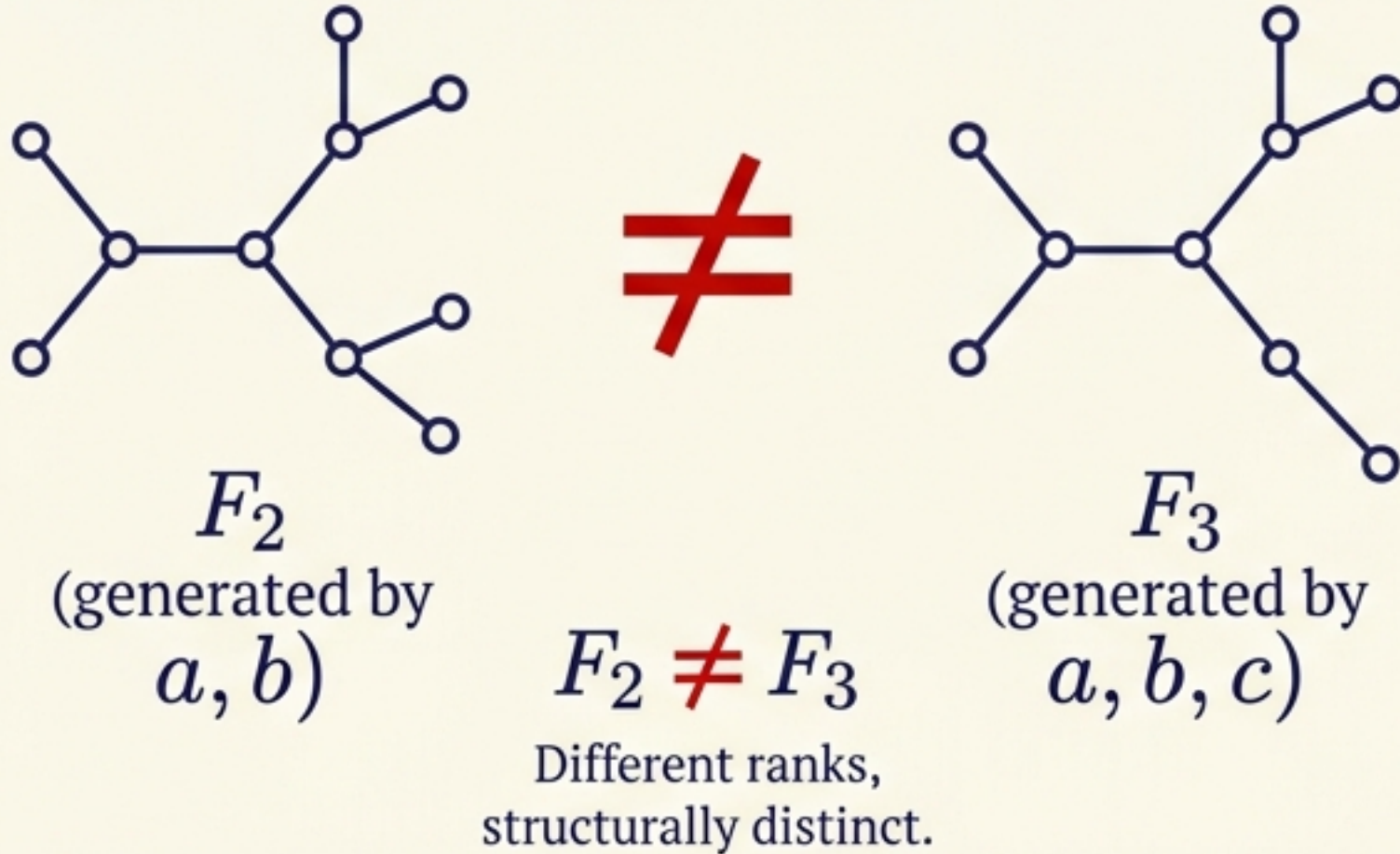




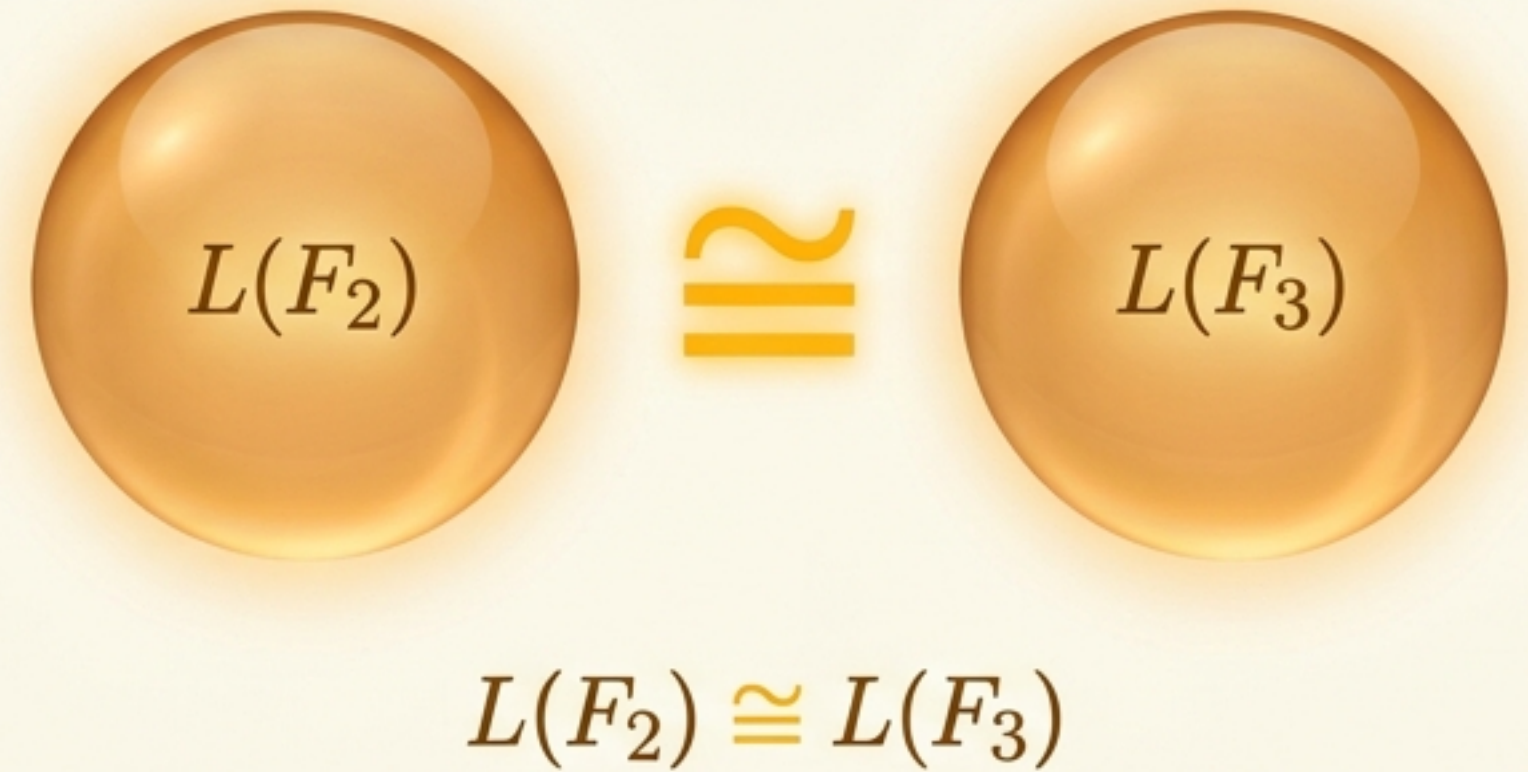
The Free Group Factor Problem: Solved

How AI and trace-preserving polynomial flows resolved an 80-year-old mystery in operator algebras.

The Discrete Groups



The Continuous Algebras



The Paradox: If we look only at the continuous symmetries of these discrete groups, do they remember how many generators they started with?

The answer is no. This presentation details the constructive proof of this ultimate equivalence.

Finite (Linear Algebra)

Infinite (Von Neumann)

Space

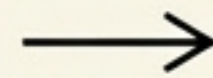
$\mathbb{C}^n$
(Finite vectors)



$\ell^2(G)$ (Square-summable functions on group G)

Operators

$n \times n$ Matrices



Bounded operators, closed in the weak-operator topology

Action

Matrix multiplication



Left translation $\lambda(g)$ as a unitary operator

Trace

Sum of diagonals divided by n

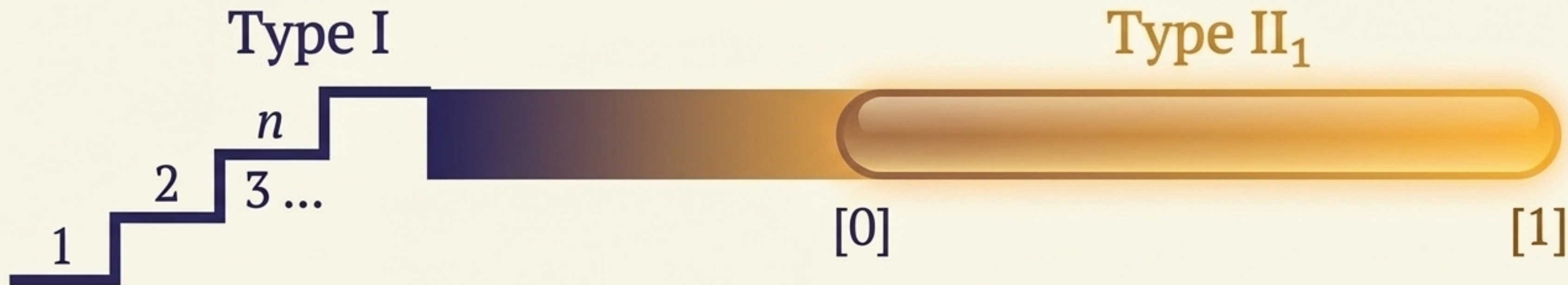


Canonical trace $\tau(x) = \langle x\delta_e, \delta_e \rangle$

A von Neumann algebra $L(G)$ is the infinite-dimensional limit of matrix mechanics, governed by a continuous trace.

What makes a II_1 Factor unique?

Spectrum of Dimensionality



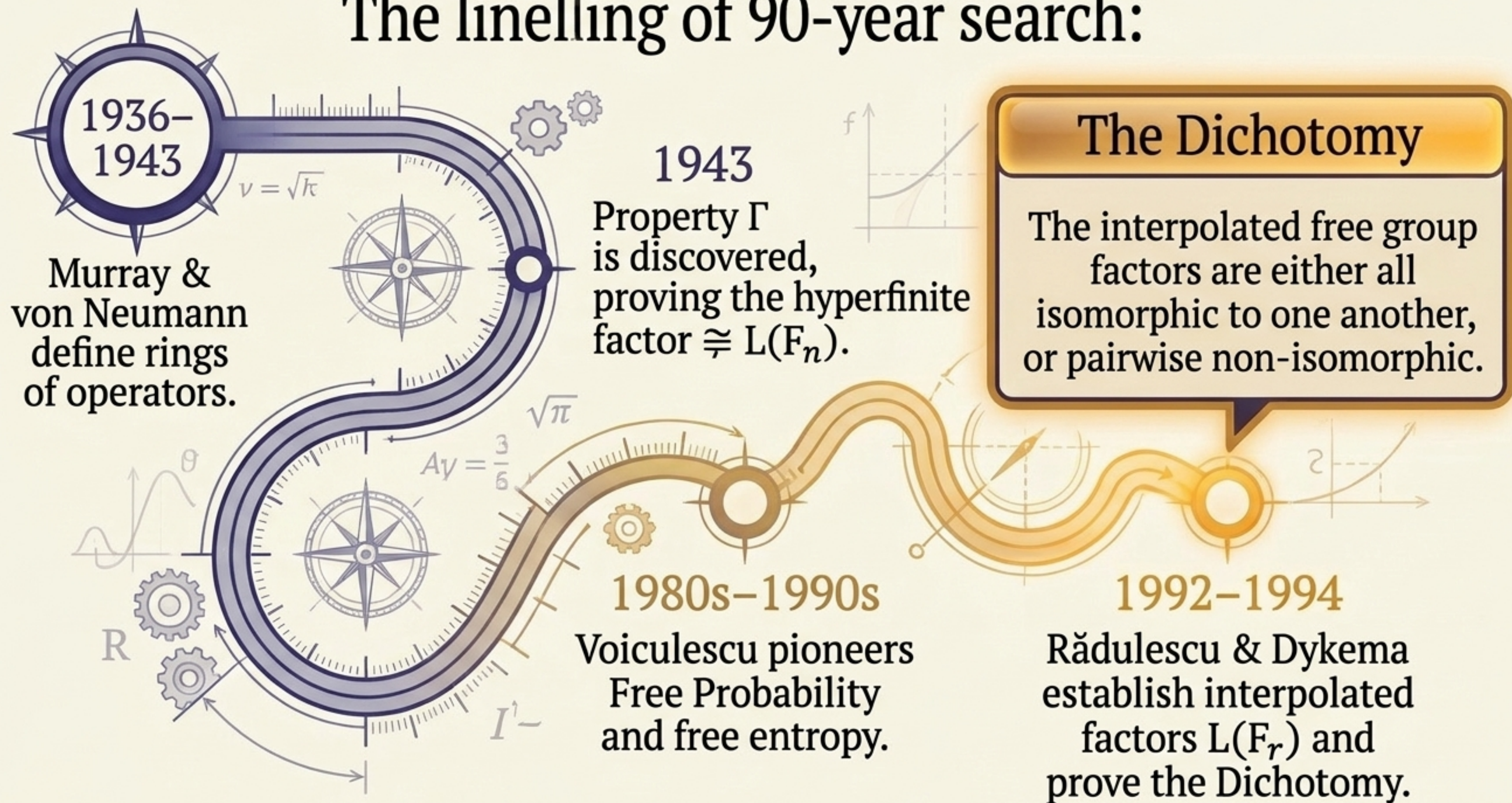
Matrix algebras.
Dimensions are strictly discrete.

Projections can be continuously halved.
A continuous geometry of dimension.

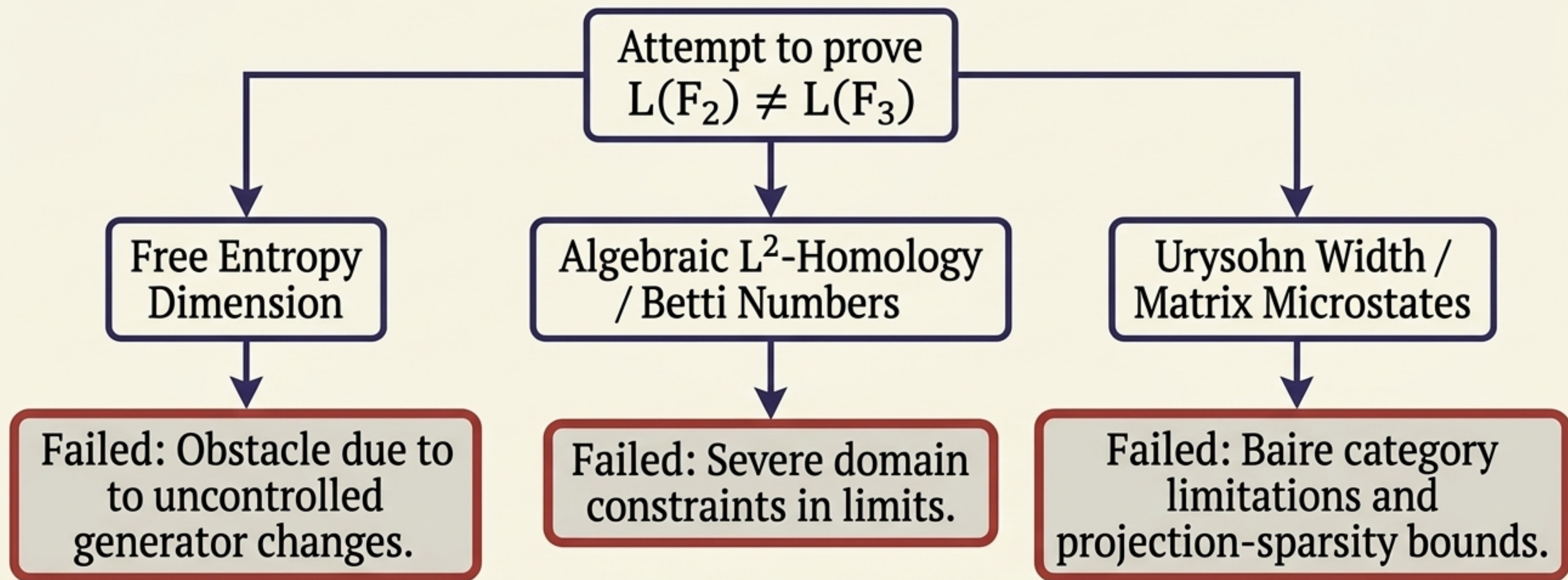
Factors are the prime numbers of von Neumann algebras—their center consists only of scalar multiples of the identity.

For any $n \geq 2$, the free group factor $L(F_n)$ is a canonical II_1 factor.

The linelling of 90-year search:



The Graveyard of Invariants



Every known invariant fails. You cannot prove they are different by searching for topological or algebraic obstructions.

Search for
Obstructions
(Prove $L(F_2) \neq L(F_3)$)

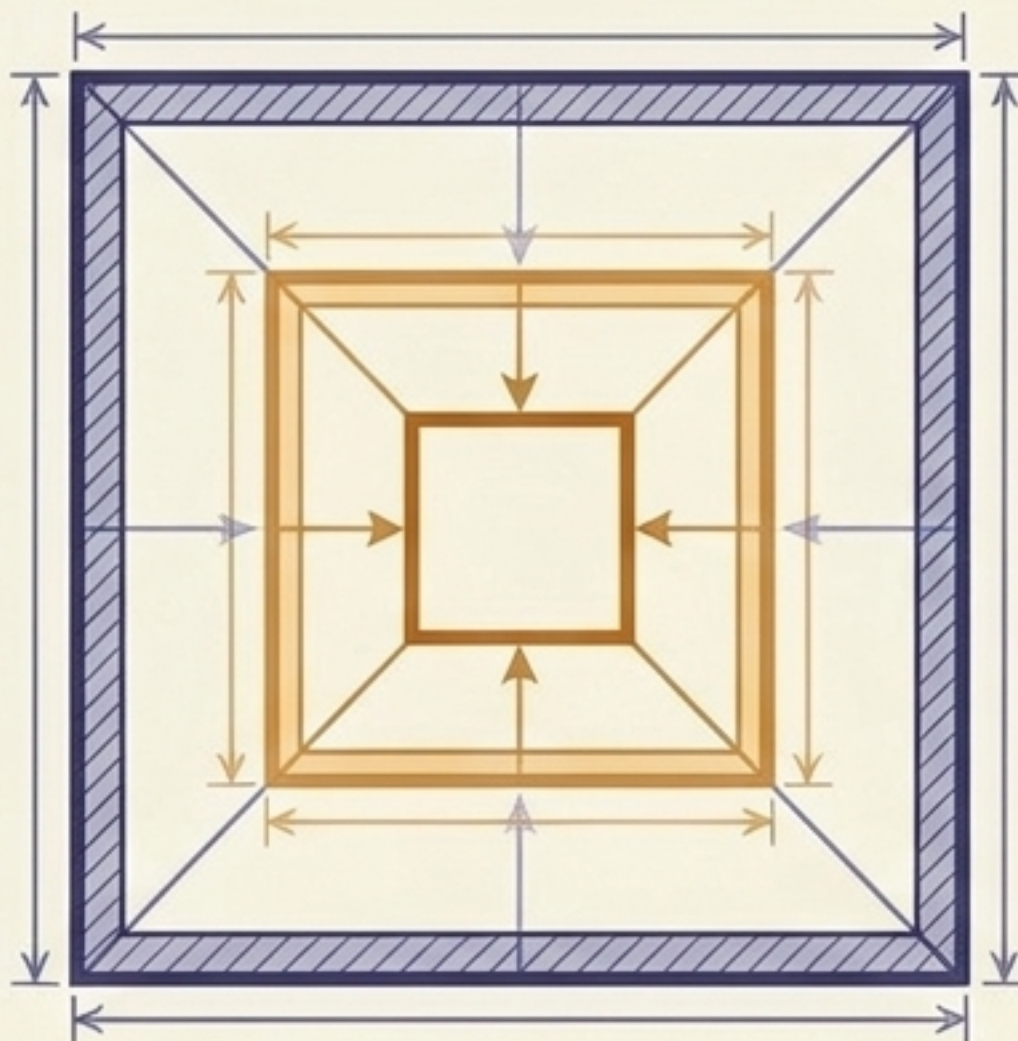
Because of the
established Dichotomy,
we only need to prove
that one single pair of
distinct finite ranks
are isomorphic.

Constructive
Isomorphism
(Prove $L(F_2) \cong L(F_3)$)

Theorem Statement

$$L(F_n) \cong L(F_{n+1}) \text{ for } n \geq 3.$$

The Amplification Calculus



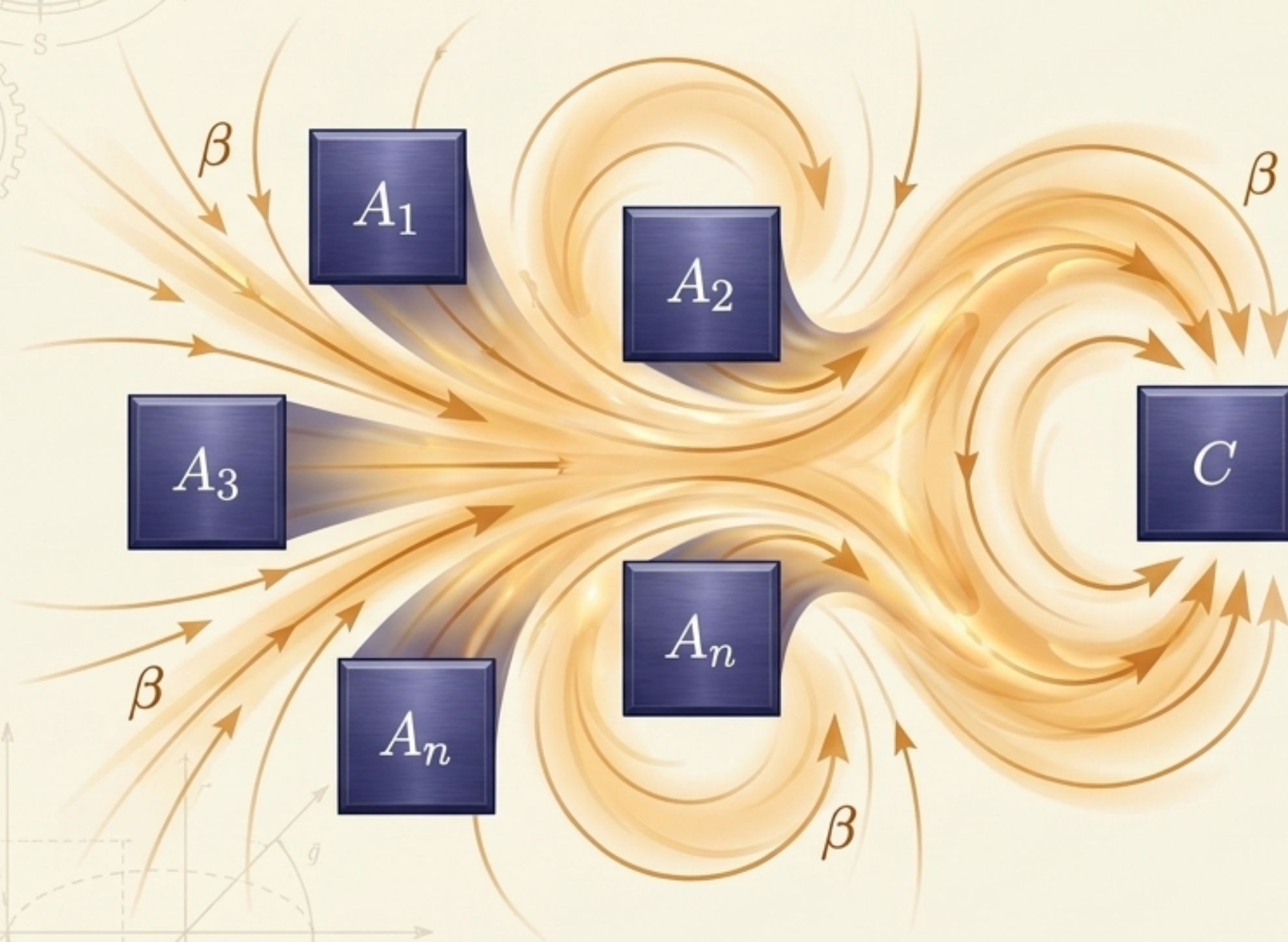
Structured Steps:

- Step 1: Theorem 1.1 establishes a chain: $L(F_3) \cong L(F_4) \cong L(F_5)$.
- Step 2: Dykema-Rădulescu Formula: $L(F_r)^t \cong L(F_{1+(r-1)/t^2})$.
- Step 3: Amplify the equivalence using $t = \sqrt{2}$.

$$L(F_3)^{\sqrt{2}} \cong L(F_2) \quad L(F_5)^{\sqrt{2}} \cong L(F_3)$$

Therefore: $L(F_2) \cong L(F_3)$.

The Proof Engine: Trace-Preserving Flows



- Start with free Haar unitaries $A_1, \dots, A_n$ and C .
- Build a trace-preserving flow β , driven by a bounded logarithm $S = \arg(C)$.
- The flow moves the A_j unitaries a tiny fraction in operator norm, while strategically moving a specific long word (A_w) close to $C \cdot A_w$.

Fox Calculus & Free Prefixes

$$a = x_1, b_j = x_2^j x_3 x_2^{-j} \quad p_j = (ab_1)(ab_2) \dots (ab_j) \quad w_m = p_m a$$

$$w_m = \underbrace{(ab_1)}_{p_1} \underbrace{(ab_1)}_{p_2} ab_2 \cdots \underbrace{ab_m}_{p_m} a$$

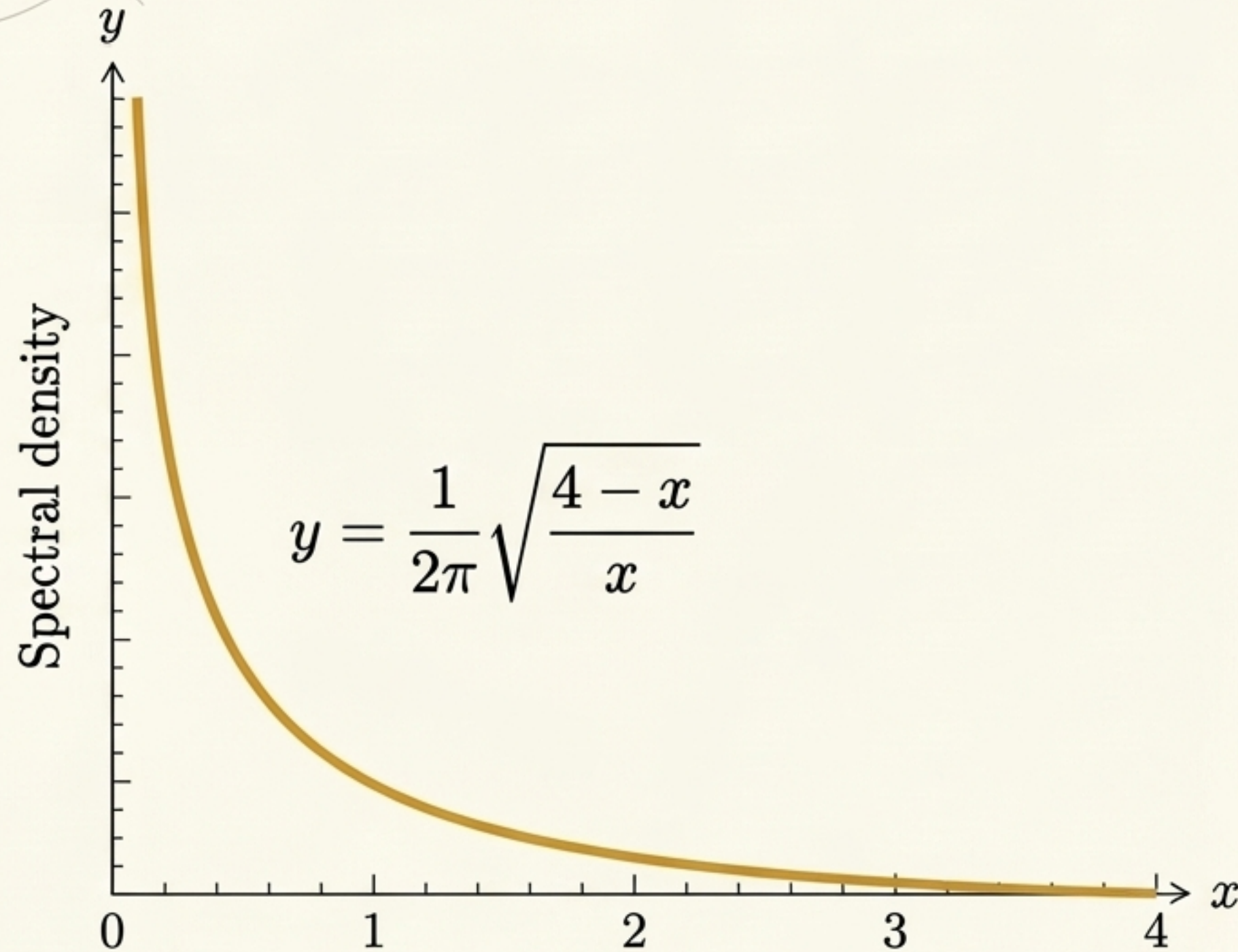
Fox Free Differential Calculus:
Successive letters in a word
contribute their generator coefficients
translated by the preceding prefix.

The coefficient of the word is
 $D_w = T_m h$, where the operator

$$T_m = 1 + \sum \lambda(p_j).$$

Only the letters a contribute, so the coefficient of the word is
a sum of shifted copies of h over the free prefixes.

Catalan Moments and No Atom at Zero

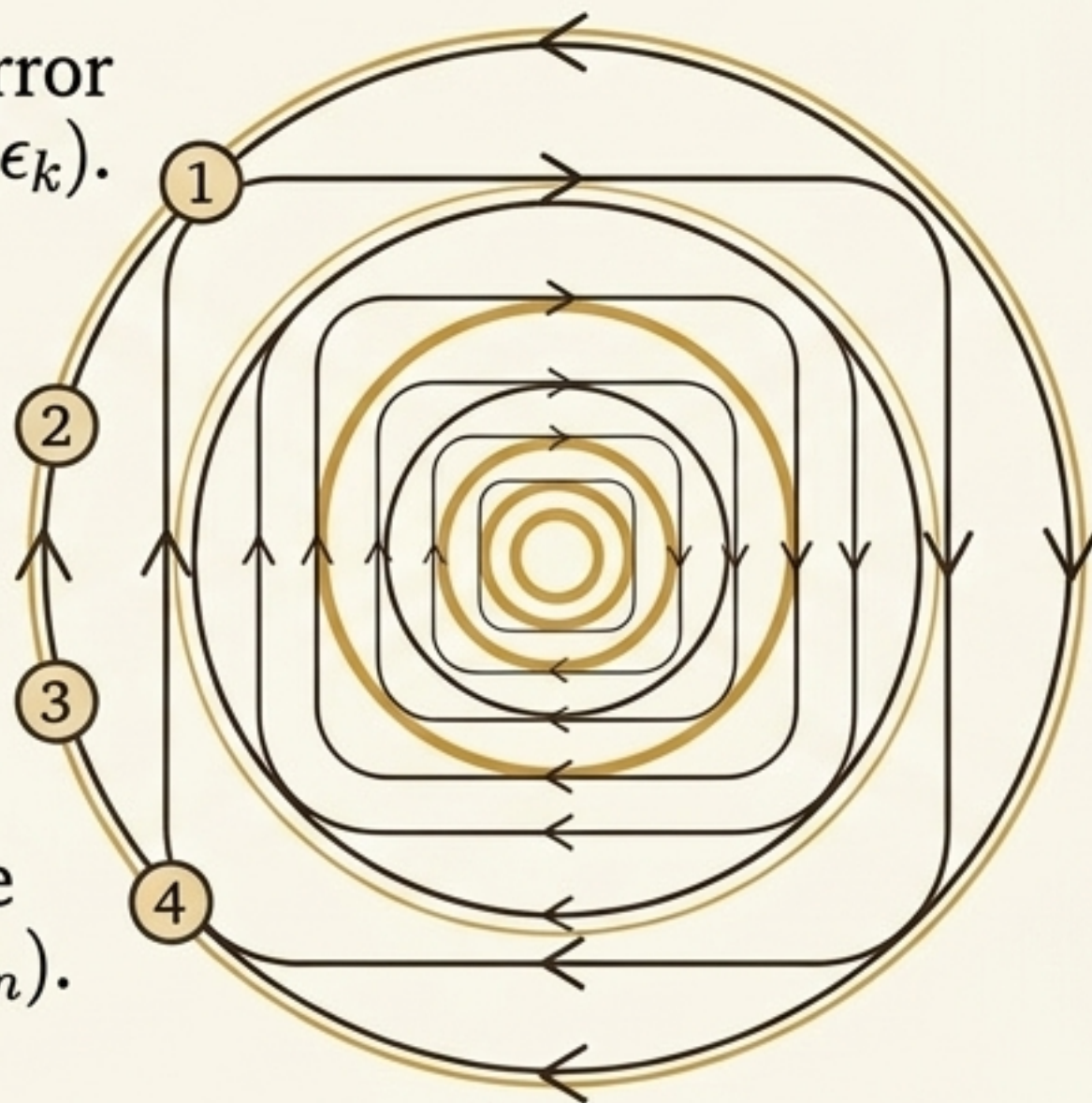


By counting Catalan moments of the free prefixes, the spectral limit yields a measure with no atom at zero.

Why this matters: The limit law has no point mass at zero, so after discarding a thin strip of small spectrum the operator can be inverted, giving a small vector h with $T_m h$ close to δ_e .

Iteration & Shrinking Error Budgets

1. Choose an error tolerance (ϵ_k).
2. Find a witness word (w_k).
3. Fix the future error budget (r_k).
4. Perturb the tuple ($A_1 \dots A_n$).

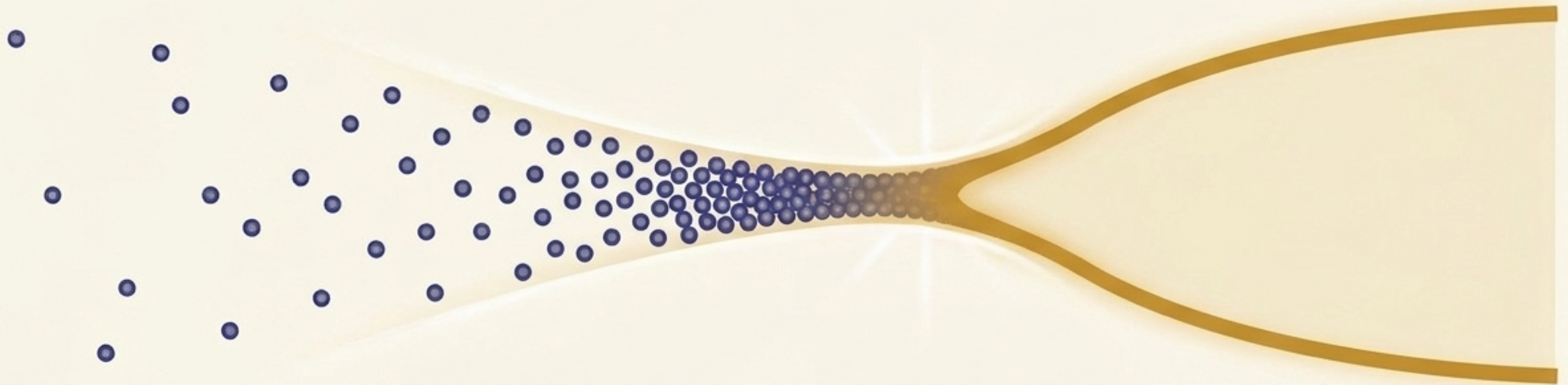


Adaptive Budgeting

The witness word is chosen before future budgets are locked in.

By repeating the automorphism with geometrically shrinking error budgets, a word built strictly from the moved A_j unitaries successfully approximates the old extra variable C .

Limiting Generation



①

The sequence of the first n coordinates converges in operator norm, persisting as a joint Haar distribution.

②

A trace-preserving conditional expectation argument proves that the word algebra of this limiting tuple is dense in $\mathcal{L}^2$.

③

Result: The limit completely generates the ambient factor $\mathcal{L}(\mathcal{F}_{n+1})$.

The Ripple Effects

Interpolated Factors

All interpolated free group factors $\mathcal{L}(\mathcal{F}_r)$ (for $1 < r \leq \infty$) are absolutely isomorphic.

Fundamental Group

The fundamental group of these factors is exactly $\mathbb{R}_+$.

Free Entropy

Voiculescu's Free Entropy Dimension (δ) is not invariant under changing von Neumann generators. (It can take any integer value $n \geq 2$ on generating tuples of $\mathcal{L}(\mathcal{F}_2)$).

The isomorphic resolution destroys the hope that free entropy dimension only depends on the generated algebra.

Mechanized Truth

```
theorem allInterpolatedIsomorphic  
  (r s : ENNReal) (hr : 1 < r) (hs : 1 < s) :  
  Nonempty (NormalTracialEquiv ...)
```

The openai/math Lean scope document says the formalization proves that all interpolated free group factors, including the infinite one, are isomorphic. The paper is an unrefereed preprint dated 23 September 2026.